General Physics · Formula & unit sheet

$g = 9.8\ m/s^2$ · $k = 9\times10^9\ N\cdot m^2/C^2$ · $\varepsilon_o = 8.85\times10^{-12}\ C^2/N\cdot m^2$ · $e = 1.6\times10^{-19}\ C$ · $c = 3\times10^8\ m/s$ · $\sigma = 5.67\times10^{-8}\ W/m^2K^4$ · $1\ cal = 4.186\ J$ · $P_o = 1.013\times10^5\ Pa$ · $T_K = T_C + 273$

1. Units and Dimension الوحدات والأبعاد

$n = 10^{-9}, \mu = 10^{-6}, m = 10^{-3}, c = 10^{-2}, d = 10^{-1}, da = 10^{1}, k = 10^{3}, M = 10^{6}, G = 10^{9}$

n 10^{-9} · **μ** 10^{-6} · **m** 10^{-3} · **k** 10^{3} · **M** 10^{6} · **G** 10^{9}

$$1\,km/h = \frac{10^3\ m}{3600\ s} = \frac{1}{3.6}\ m/s$$

$$1\,g/cm^3 = \frac{10^{-3}\ kg}{(10^{-2})^3\ m^3} = 10^3\ kg/m^3$$

ρ kg/m³

$$[F] = [m][a] = M\cdot LT^{-2} = MLT^{-2}$$

2. Vectors المتجهات

$$C = A - B = A + (-B)$$

$$C = \sqrt{(A^2 + B^2)}\ ,\ \theta = \tan^{-1}\left(\frac{B}{A}\right)$$

θ °

$$C = \sqrt{(A^2 + B^2 - 2AB\cos\theta)}$$

θ °

$$A_x = A\cos\theta\ ,\ A_y = A\sin\theta$$

θ °

$$A + B = (A_x{+}B_x)\hat{\imath} + (A_y{+}B_y)\hat{\jmath} + (A_z{+}B_z)\hat{k}$$

$$A\cdot B = AB\cos\theta = A_xB_x + A_y\,B_y + A_z\,B_z$$

θ °

$$|A\times B| = AB\sin\theta$$

θ °

3. Motion in one dimension الحركة في بُعد واحد

$$\Delta x = x_2 - x_1$$

Δx m · **x_1, x_2** m

$$v_{av} = \frac{\Delta x}{\Delta t}\ ,\ v = \frac{dx}{dt}$$

v_{av} m/s · **v** m/s · **Δx** m · **t** s

$$a_{av} = \frac{\Delta v}{\Delta t}\ ,\ a = \frac{dv}{dt}$$

a_{av} m/s² · **a** m/s² · **Δv** m/s · **t** s

$$v_f = v_i + at$$

v_i, v_f m/s · **a** m/s² · **t** s

$$x_f = x_i + v_i\,t + \tfrac{1}{2}at^2$$

x_i, x_f m · **v_i** m/s · **a** m/s² · **t** s

$$v_f^{\,2} = v_i^{\,2} + 2a\Delta x$$

v_i, v_f m/s · **a** m/s² · **Δx** m

4. Laws of Motion قوانين الحركة

$$\Sigma F = 0 \Leftrightarrow \Sigma F_x = 0 \text{ and } \Sigma F_y = 0$$

ΣF N · **ΣF_x, ΣF_y** N

$$\Sigma F = m\,a$$

ΣF N · **m** kg · **a** m/s²

$$F_w = m\,g$$

F_w N · **m** kg · **g** m/s²

$$F_{12} = -F_{21}$$

F_{12} N · **F_{21}** N

5. Work and Energy الشغل والطاقة

$$W = F\cdot\Delta r = F\,\Delta r\cos\theta$$

W J · **F** N · **Δr** m · **θ** °

$$KE = \tfrac{1}{2}\,m\,v^2\ ,\ W = KE_f - KE_i$$

KE J · **m** kg · **v** m/s · **W** J

$$\Delta PE = m\,g\,\Delta h$$

ΔPE J · **m** kg · **g** m/s² · **Δh** m

$$W = \Delta KE + \Delta PE$$

W J · **ΔKE** J · **ΔPE** J

$$P = \frac{dW}{dt}\ (\text{average: } P = \frac{W}{t})$$

P W · **W** J · **t** s

6. Elasticity and Fluid Mechanics المرونة وميكانيكا الموائع

$$\frac{F}{A} = Y\,\frac{\Delta L}{L}$$

F N · **A** m² · **Y** N/m² · **ΔL** m · **L** m

$$\frac{F}{A} = -B\,\frac{\Delta V}{V}$$

F/A N/m² · **B** N/m² · **ΔV** m³ · **V** m³

$$\frac{F}{A} = S\,\frac{x}{L}$$

F N · **A** m² · **S** N/m² · **x** m · **L** m

$$\rho = \frac{m}{V}$$

ρ kg/m³ · **m** kg · **V** m³

$$P = \frac{F}{A}\ ,\ P = P_0 + \rho gh$$

P Pa · **F** N · **A** m² · **P_o** Pa · **ρ** kg/m³ · **h** m

$$\frac{f}{a} = \frac{F}{A}$$

f, F N · **a, A** m²

$$P_0 = \rho gh\ ,\ P = P_0 + \rho gh$$

P_o Pa · **ρ** kg/m³ · **h** m

$$A_1v_1 = A_2v_2$$

A m² · **v** m/s

$$P_1 + h_1\rho g + \tfrac{1}{2}\rho v_1^{\,2} = P_2 + h_2\rho g + \tfrac{1}{2}\rho v_2^{\,2}$$

P Pa · **h** m · **ρ** kg/m³ · **v** m/s

7. Heat الحرارة

$$T_C = \frac{5}{9}\,(T_F - 32)\ ,\ T_K = T_C + 273$$

T_C °C · **T_F** °F · **T_K** K

$$\Delta\ell = \alpha\,\ell\,\Delta T\ ,\ \Delta S = \gamma\,S\,\Delta T\ ,\ \Delta V = \beta\,V\,\Delta T$$

Δℓ, ΔS, ΔV m, m², m³ · **α** °C⁻¹ or K⁻¹ · **γ, β** °C⁻¹ · **ΔT** °C or K

$$Q = c\,m\,\Delta T$$

Q J · **c** J/kg·K · **m** kg · **ΔT** K or °C

$$Q = m\,L$$

Q J · **m** kg · **L** J/kg

$$\Sigma\,Q_{lost} = \Sigma\,Q_{gained}$$

Q J or cal · **T_f** °C

$$H = \frac{k_c\,A\,(T_1 - T_2)}{x}$$

H W or cal/s · **k_c** W/m·K · **A** m² · **$T_1 - T_2$** K or °C · **x** m

$$Q = \sigma\,A\,T^4$$

Q W · **σ** W/m²K⁴ · **A** m² · **T** K

8. Static Electricity الكهرباء الساكنة

$$F = \frac{k_e\,|q_1\,q_2|}{r^2}$$

F N · **k_e** N·m²/C² · **q_1, q_2** C · **r** m

$$E = \frac{F}{q_t} = \frac{k_e\,q}{r^2}$$

E N/C · **F** N · **q_t** C · **q** C · **r** m

$$\Delta V = \frac{W}{q_t} = E\,d$$

ΔV V · **W** J · **q_t** C · **E** N/C = V/m · **d** m

$$V = \frac{k_e\,q}{r}$$

V V · **q** C · **r** m

$$Q = C\,V\ ,\ C = \frac{\varepsilon_0\,A}{d}$$

Q C · **C** F · **V** V · **ε_o** C²/N·m² · **A** m² · **d** m

$$\text{Series: } \frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \ldots \cdot \text{Parallel: } C_{eq} = C_1 + C_2 + \ldots$$

C_{eq} F · **C_1, C_2** F

$$U = \tfrac{1}{2}\,C\,V^2 = \frac{Q^2}{2C} = \tfrac{1}{2}\,Q\,V$$

U J · **C** F · **V** V · **Q** C

9. Current and Resistance التيار والمقاومة

$$I = \frac{dq}{dt} = n\,e\,v\,A$$

I A · **q** C · **t** s · **n** m⁻³ · **e** C · **v** m/s · **A** m²

$$J = \frac{I}{A} = n\,e\,v$$

J A/m² · **I** A · **A** m²

$$V = R\,I$$

V V · **R** Ω · **I** A

$$R = \frac{\rho\,\ell}{A}\ ,\ \rho = \frac{1}{\sigma}$$

R Ω · **ρ** Ω·m · **ℓ** m · **A** m² · **σ** $(\Omega\cdot m)^{-1}$

$$P = V\,I = R\,I^2 = \frac{V^2}{R}$$

P W · **V** V · **I** A · **R** Ω

$$\text{Series: } R_{eq} = R_1 + R_2 + \ldots \cdot \text{Parallel: } \frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \ldots$$

R_{eq} Ω · **R_1, R_2** Ω

10. Light الضوء

$$\theta_1 = \theta_2$$

θ_1 ° · **θ_2** °

$$n = \frac{c}{v} = \frac{\lambda}{\lambda_n}$$

c m/s · **v** m/s · **λ, λ_n** m

$$n_1\sin\theta_1 = n_2\sin\theta_2$$

θ_1, θ_2 °

$$\sin\theta_c = \frac{n_2}{n_1}$$

θ_c °

Units: SI unless stated. Δ = final − initial. Angles in Snell's law and work are measured as defined in each lesson. Physics Learning Lab · smalgamdi-gif.github.io/physic